Anatomy of an Excellent Engineering Prompt

Each component serves a specific purpose in guiding AI response quality

Role Assignment	"Act as a senior electromechanical engineer specializing in automotive systems"	Sets expertise level and perspective for response
Context	"I am designing a regenerative braking system for an electric vehicle"	Establishes the problem domain and application
Specifications	"Vehicle mass: 2000kg, max speed: 150km/h, battery: 400V nominal"	Provides concrete parameters for calculations
Constraints	"Must comply with ISO 26262 ASIL−B, budget under $500/unit"	Defines boundaries and requirements
Deliverables	"Provide: 1) Energy calculations, 2) Component selection, 3) Control strategy"	Lists exactly what you need from the AI
Format	"Format as a technical design document with diagrams described"	Ensures output is immediately usable
Standards	"Reference ISO 26262 and SAE J2954 where applicable"	Grounds response in industry requirements